

Access RDS DB deployed in Private server using MySQL Workbench

After - RDS DB deployed in Private server using MySQL Workbench.

1. MySQL client – we can manage the RDS database via Mysql client in the Bastion host.
2. MySQL Server – used to create and manage database in the server . it is the server it self.

Now install the mysql agent in Bastion host and manage the AWS RDS DB

Install the MySQL client in bastion host .

sudo yum install mariadb105 -y

```
[ec2-user@ip-10-0-9-245 ~]$ sudo yum install mariadb105 -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
462 kB/s | 61 kB    00:00
=====
Package                        Architecture  Version                                Repository    Size
=====
Installing:
mariadb105                    x86_64        3:10.5.29-1.amzn2023.0.1              amazonlinux   1.5 M
Installing dependencies:
mariadb-connector-c            x86_64        3.3.10-1.amzn2023.0.2                  amazonlinux   211 k
mariadb-connector-c-config     noarch        3.3.10-1.amzn2023.0.2                  amazonlinux   9.3 k
mariadb105-common              x86_64        3:10.5.29-1.amzn2023.0.1              amazonlinux   28 k
perl-Sys-Hostname               x86_64        1.23-477.amzn2023.0.9                  amazonlinux   17 k
Transaction Summary
=====
Install      1 Package
Total size:  1.5 M
Total download size: 1.5 M
Estimated time: 0.1 s
```

Now connect with RDS DB ec2 instance via **endpoint DNS** , which you want to manage.

#mysql -h <endpoint of RDS DB > -u admin -p

[-h > hostname which you want to connect , -u > admin , -p > password]

#mysql -h database-1.c6xkgac6c5dq.us-east-1.rds.amazonaws.com -u admin -p

```
[ec2-user@ip-10-0-9-245 ~]$ mysql -h database-1.c6xkgac6c5dq.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 397
Server version: 8.4.9 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| test |
+-----+
5 rows in set (0.007 sec)

MySQL [(none)]>
```

As you can see we are managing RDS database in our Bastion host using MySQL client.

But the actual database is still in the AWS RDS private EC2 instance .

Now install the MySQL server in the Bastion host and manage the database .

Remove the MySQL agent - # **sudo yum remove mariadb105 -y**

```
[ec2-user@ip-10-0-9-245 ~]$ sudo yum remove mariadb105 -y
Dependencies resolved.
=====
Package                                Architecture    Version                                Repository      Size
=====
Removing:
mariadb105                             x86_64          3:10.5.29-1.amzn2023.0.1             @amazonlinux    18 M
Removing unused dependencies:
mariadb-connector-c                     x86_64          3.3.10-1.amzn2023.0.2                 @amazonlinux    584 k
mariadb-connector-c-config              noarch          3.3.10-1.amzn2023.0.2                 @amazonlinux    497
mariadb105-common                       x86_64          3:10.5.29-1.amzn2023.0.1             @amazonlinux    179 k
perl-Sys-Hostname                       x86_64          1.23-477.amzn2023.0.9                 @amazonlinux    21 k
Transaction Summary
=====
Remove: 5 Packages
```

----- DB on ec2-----

#"Installing MySQL server on Amazon Linux 2023"

sudo dnf update

sudo wget https://dev.mysql.com/get/mysql80-community-release-el9-4.noarch.rpm

sudo dnf install mysql80-community-release-el9-4.noarch.rpm

sudo dnf install mysql-community-server

```
Installed:
mysql-community-client-8.0.46-1.el9.x86_64      mysql-community-client-plugins-8.0.46-1.el9.x86_64
mysql-community-common-8.0.46-1.el9.x86_64      mysql-community-icu-data-files-8.0.46-1.el9.x86_64
mysql-community-libs-8.0.46-1.el9.x86_64        mysql-community-server-8.0.46-1.el9.x86_64

Complete!
[ec2-user@ip-10-0-9-245 ~]$
```

mysql -V

sudo systemctl start mysqld #Starts MySQL service

```
mysql Ver 8.0.46 for Linux on x86_64 (MySQL Community Server - GPL)
[ec2-user@ip-10-0-9-245 ~]$ sudo systemctl start mysqld
[ec2-user@ip-10-0-9-245 ~]$ ss -tlnl
Netid State Recv-Q Send-Q Local Address:Port Peer Address:Port
udp UNCONN 0 0 127.0.0.1:323 0.0.0.0:*
udp UNCONN 0 0 10.0.9.245%ens5:68 0.0.0.0:*
udp UNCONN 0 0 [:::]:323 [:::]*
udp UNCONN 0 0 [fe80::b5:f5ff:fe6:6d07]%ens5:546 [:::]*
tcp LISTEN 0 128 0.0.0.0:22 0.0.0.0:*
tcp LISTEN 0 128 [:::]:22 [:::]*
tcp LISTEN 0 151 *:3306 *:
tcp LISTEN 0 70 *:33060 *:
[ec2-user@ip-10-0-9-245 ~]$
```

sudo systemctl enable mysqld #Enabled MySQL service to restart on bot

systemctl status mysqld #Check MySQL service running status

#Get Initial Password

```
sudo grep 'temporary password' /var/log/mysql.log
```

(note temp passwd 3F3ot=7b!lg)

connect MySQL and run below command

```
mysql -u root -p
```

give temp password latter you can run below command for custm password

```
ALTER USER 'root'@'localhost' IDENTIFIED BY 'Passwd@12';
```

```
FLUSH PRIVILEGES;
```

```
mysql -u root -p
```

#create sample data base

```
CREATE DATABASE IF NOT EXISTS demodb;
```

```
SHOW DATABASES;
```

```
CREATE TABLE demodb.persons (
```

```
    PersonID int,
```

```
    LastName varchar(255),
```

```
    FirstName varchar(255),
```

```
    Address varchar(255),
```

```
    City varchar(255)
```

```
);
```

```

mysql> create database demodb;
Query OK, 1 row affected (0.00 sec)

mysql> CREATE TABLE demodb.persons (
  ->   PersonID int,
  ->   LastName varchar(255),
  ->   FirstName varchar(255),
  ->   Address varchar(255),
5)
  ->   City varchar(255)
  -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> show databases;
+-----+
| Database |
+-----+
| demodb   |
| information_schema |
| mysql    |
| performance_schema |
| sys      |
+-----+
5 rows in set (0.00 sec)

mysql>

```

USE DBName;

give demodb

SHOW TABLES;

```

mysql> use demodb;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+-----+
| Tables_in_demodb |
+-----+
| persons          |
+-----+
1 row in set (0.00 sec)

mysql>

```